Candidate's name:……………………………………...……………………………......

Signature:…………..**COMBINATION:**………........

P525/2
CHEMISTRY
Paper 2
(Practical)
July/Aug.2025
2 hours

THE CHEMISTRY DEPARTMENT

CHEMISTRY

S5

Paper 2
(Practical)

2 hours

INSTRUCTIONS TO CANDIDATES:

This paper consists of **one compulsory** *examination item.*

Answers to this item are to be written in the spaces provided in this paper. Use **blue** *or* **black** *ink. Any work done in pencil* **except** *drawings and graph will* **not** *be scored.*

All *working* ***must*** *be clearly shown. Graph paper will be provided.*

Mathematical tables and silent non-programmable scientific calculators may be used.

You are **not** *allowed to use reference books.*

Candidates are advised to carefully read the item, make sure they have all the apparatus and chemicals they may need and then **plan** *appropriately before starting.*

TURNOVER

ITEM

A school chemistry lab uses a sodium hydroxide solution of concentration at approximately 40 g/dm^3 for acid-qualitative analysis. Over time, the solution absorbs carbon dioxide from air, forming sodium carbonate and reducing its concentration. The lab technician suspects the current solution may no longer meet the required concentration.

Accurate concentration is critical for an upcoming test on unknown carboxylic acids. An incorrect sodium hydroxide concentration could lead to faulty interpretations.

To verify its suitability, there is need to standardise the sodium hydroxide solution using a primary standard solution of oxalic acid ($H_2C_2O_4 \cdot 2H_2O$), prepared by dissolving 6.90–7.10 g in 250 cm^3 of distilled water.

You will then determine whether the sodium hydroxide solution's concentration is approximately 40g per litre and advise whether it can be used as-is, or must be discarded. The outcome will ensure the lab's acid-base experiments and the students' results remain accurate and reliable.

Sodium hydroxide reacts with oxalic acid according to the following equation;

$$H_2C_2O_4(aq) + 2NaOH(aq) \longrightarrow Na_2C_2O_4(aq) + 2H_2O(l)$$

You are provided with the following;

- 7.5 g of Oxalic acid ($H_2C_2O_4 \cdot 2H_2O$)
- distilled water, sufficient for solution preparation and rinsing
- 100cm^3 of GA1 which is the solution of sodium hydroxide used by the laboratory.
- phenolphthalein indicator
- 250.0 cm^3 volumetric flask
- Other laboratory apparatus suitable for a titration experiment.

Task;

As a chemistry learner;

(a) Design an experiment that you will carry out .***(Your design should include; aim, hypothesis, variables, apparatus and materials, procedure, risks and their mitigations)***

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(b) Carry out the experiment and record your results.

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(c) Interprete and analyse your results.

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(d) What conclusion and recommendation can you give basing on your results?

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END